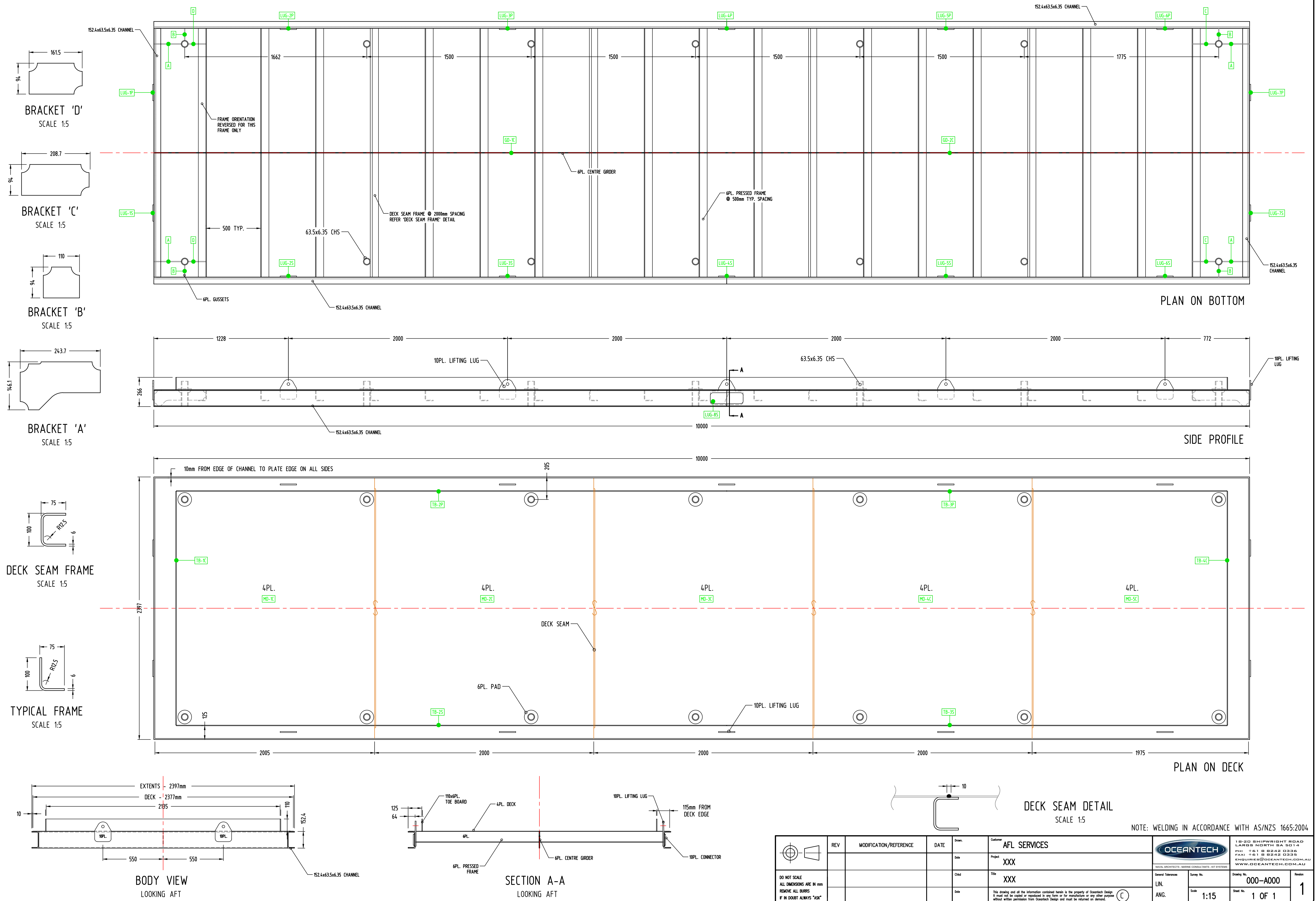
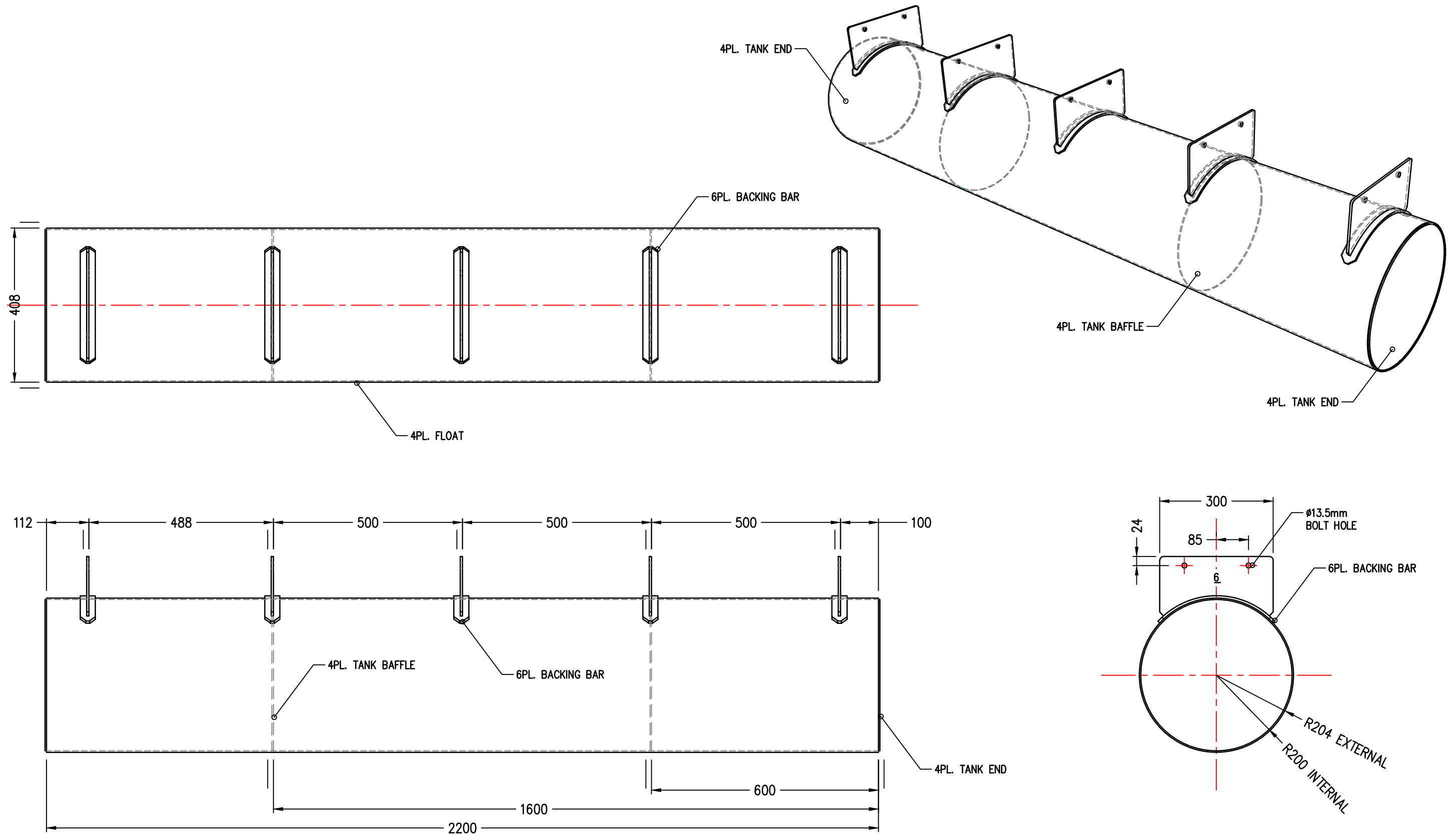
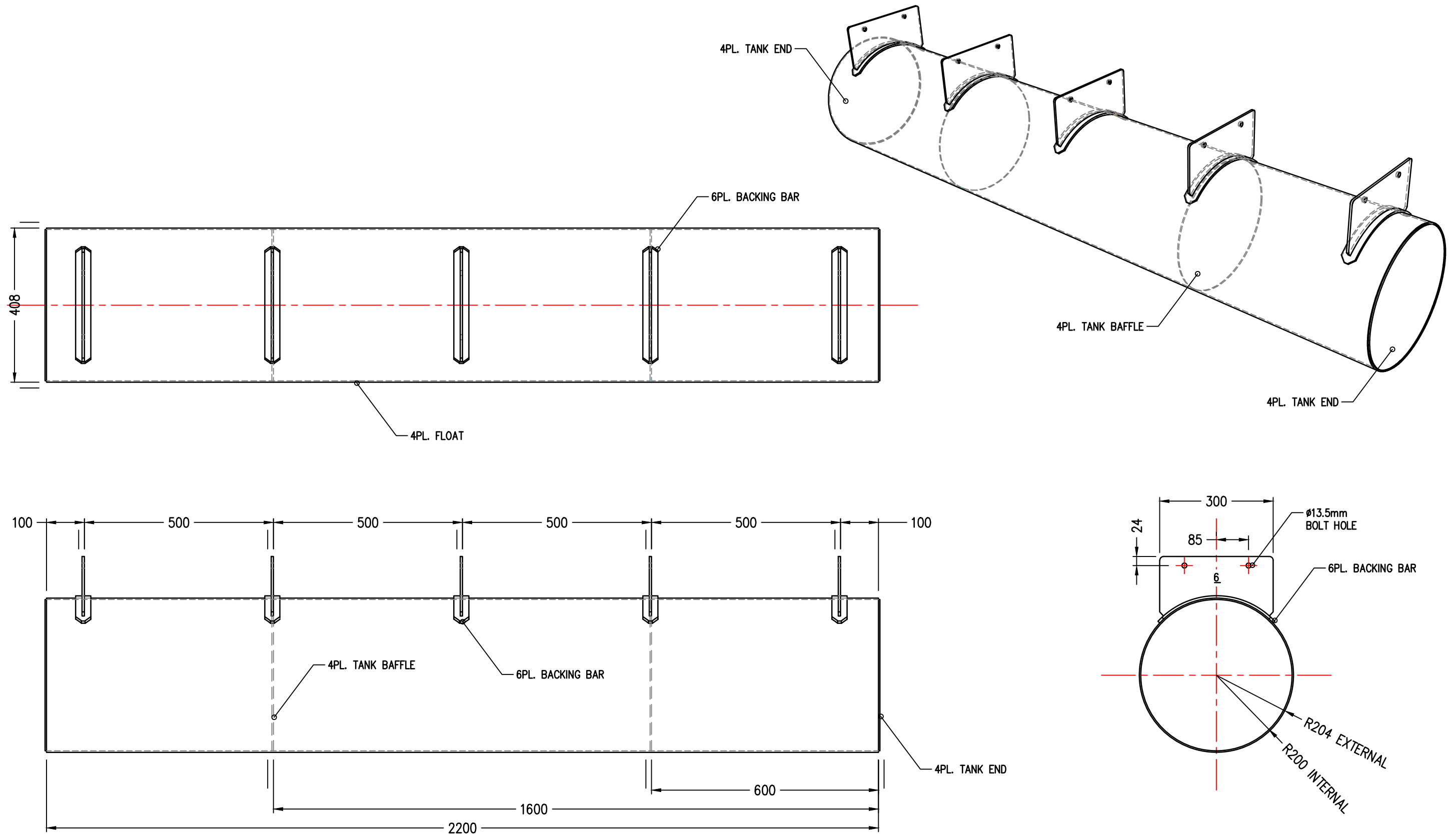






QTY REQUIRED: 2 PER PONTOON

	REV	MODIFICATION/REFERENCE	DATE	Drawn M.T	Customer AFL SERVICES		18-20 SHIPWRIGHT ROAD LARGES NORTH SA 5014 PH: +61 8 8242 0336 FAX: +61 8 8242 0335 ENQUIRIES@OCEANTECH.COM.AU WWW.OCEANTECH.COM.AU		
				Date 20.03.20	Project				
DO NOT SCALE ALL DIMENSIONS ARE IN mm REMOVE ALL BURRS IF IN DOUBT ALWAYS "ASK"				Chkd	Title PONTOON FLOAT CONSTRUCTION	General Tolerances LIN. ANG.	Survey No.	Drawing No. 000-A000	Revision A
				Date	This drawing and all the information contained herein is the property of Oceantech Design. It must not be copied or reproduced in any form or for manufacture or any other purpose without written permission from Oceantech Design and must be returned on demand. ©		Scale 1:10	Sheet No. 1 OF 1	



QTY REQUIRED: 4 PER PONTOON

	REV	MODIFICATION/REFERENCE	DATE	Drawn M.T	Customer AFL SERVICES	 NAVAL ARCHITECTS - MARINE CONSULTANTS - KIT SYSTEMS	18-20 SHIPWRIGHT ROAD LARGES NORTH SA 5014 PH: +61 8 8242 0336 FAX: +61 8 8242 0335 ENQUIRIES@OCEANTECH.COM.AU WWW.OCEANTECH.COM.AU		
				Date 20.03.20	Project				
DO NOT SCALE ALL DIMENSIONS ARE IN mm REMOVE ALL BURRS IF IN DOUBT ALWAYS "ASK"				Chkd	Title PONTOON FLOAT CONSTRUCTION	General Tolerances LIN. ANG.	Survey No.	Drawing No. 000-A000	Revision A
				Date	This drawing and all the information contained herein is the property of Oceantech Design. It must not be copied or reproduced in any form or for manufacture or any other purpose without written permission from Oceantech Design and must be returned on demand.		Scale 1:10	Sheet No. 1 OF 1	

along the contact surface between the bearing and the end eye.
The tension shall be calculated according to the following:

$$\sigma_t = \frac{F}{T \times (D - d)} \sqrt{\frac{D^2}{d^2} - \frac{D}{d} + 1} \leq \sigma_y$$

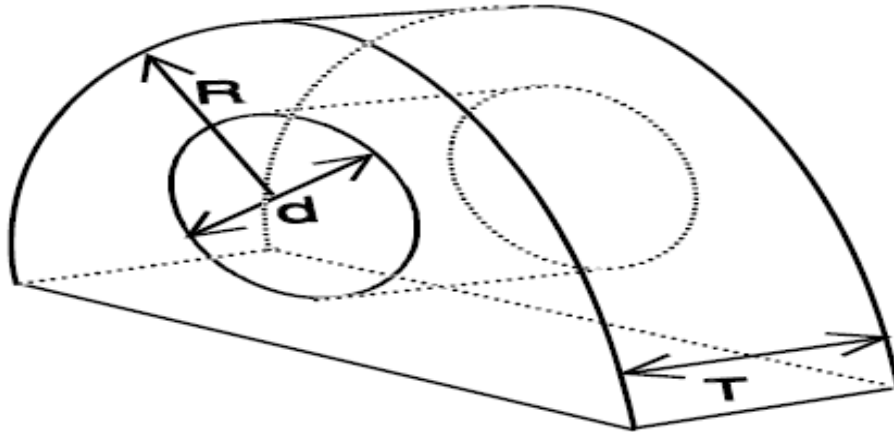


Figure 3-5
End eye

where:

F = cylinder traction force, N

T = width of end eye, mm

D = outer diameter of end eye = 2 R in fig. 5, mm

d = inner diameter of end eye (outside bearing), mm

σ_y = yield strength of the end eye material, N/mm².

5083-H116 Plate

F = 12250 N

T = 10 mm

D = 80 mm

d = 30 mm

σ_y = 215 N/mm²

σ_t = 57.17 N/mm²

Safety Factor = **3.761**

Welding Schedule

1. General

Welded connections should be joined by butt, fillet, plug or slot welds, or a combination of these. Any additional detail regarding welded connections should be sought from AS 1665, WELDING OF ALUMINIUM STRUCTURES.

Welded joints are to be detailed such that crevices or inaccessible pockets capable of retaining dirt or moisture are avoided. Where cavities are unavoidable, they are to be sealed by welding or protective compounds or made accessible for inspection and maintenance.

2. Materials

All structural plate grade to be 5083 H321/116 and structural extrusion grade to be 6082 T6 or 6061 T6 All structural materials to be LR or DNV Certified

3. Welding Processes to be used

PULSE GMAW (MIG)

4. Weld Quality Level

AN/NZS 1665:2004

5. Welding Consumables

Special care is to be taken in the distribution, storage and handling of all welding consumables. Aluminium filler metals are to be kept in a heated dry storage area with a relatively uniform temperature. Condensation on the metal surface during storage and use is to be avoided. Other welding consumables such as bare wire and welding studs are to be stored under dry conditions to prevent corrosion. Effective facilities for protecting Consumables are to be provided close to working areas.

Wire Grade	Alloy 5356
Wire Dia	1.2mm
Shielding Gas	100% Welding Grade Argon

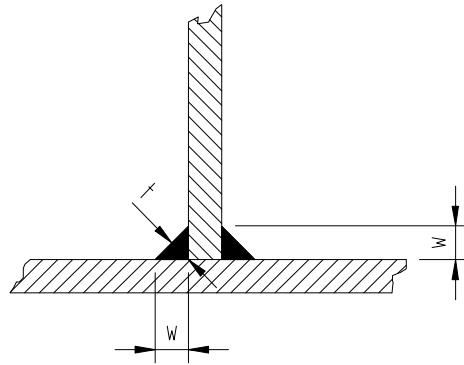
TABLE A1 - TYPE & SIZE OF WELD FOR VARIOUS STRUCTURAL CONNECTIONS

Minimum Thickness of Members	<= 5mm	>5 <= 6.5mm	>6.5 <= 8mm	>8 <= 9.5mm
Nominal Length of Fillet Weld	60	75	80	80
Nominal Leg length of Fillet Weld (w)	3	6.5	6.5	6.5
Nominal Throat Distance of Fillet Weld (t)	2	3.5	4.5	4.5
Structural Connections	Spacing of Welds			
Longitudinal Beams to Transverse Beams	Double Continuous			
Centreline Girder to Transverse Beams	Double Continuous			
Bottom Frames to Longitudinal Beams	Double Continuous			
Bottom Frames to Centreline Girder	Double Continuous			
Bottom Frames to Deck	200±	200±	225±	200±
Centreline Girder to Deck	200±	200±	225±	200±
Kick Rail to Deck	250±	250±	275±	250±
Peripheries of Deck to Longitudinal/Transverse Beams	250±	250±	275±	250±
Gussets to Deck, Beams & Frames	Double Continuous			
Doubler Pads & Lifting Lugs to Beams, Decks & Frames	Fully Welded			

Slot Weld Dimensions

	Thickness of part containing Slot Weld (millimetres)			
	4	5	6	8
w (mm)	14	15	16	18
L (mm)	56	60	64	72
S (mm)	200	200	200	200
R (mm)	7	7.5	8	9

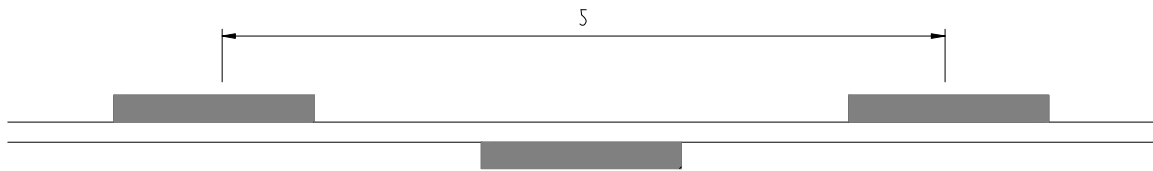
FIG. 1



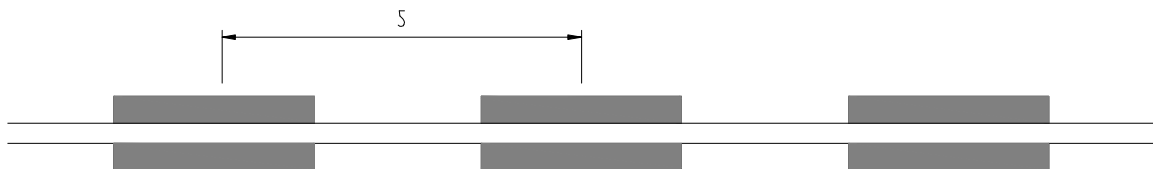
WELD DIMENSIONS

w = LEG LENGTH OF WELD

t = THROAT DISTANCE OF WELD



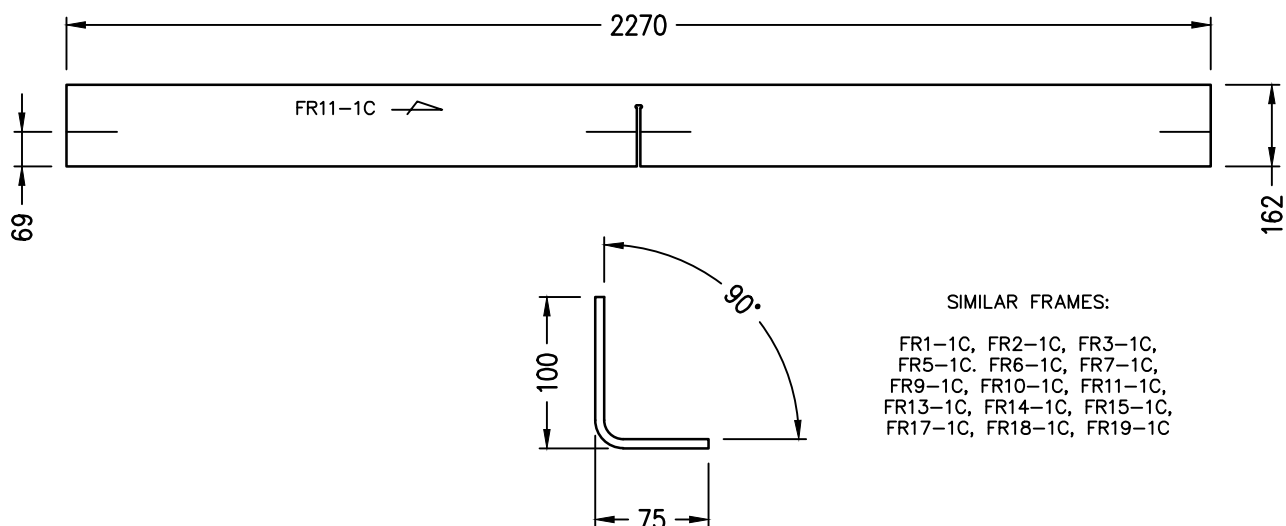
STAGGERED INTERMITTENT FILLET WELD



CHAIN INTERMITTENT FILLET WELD

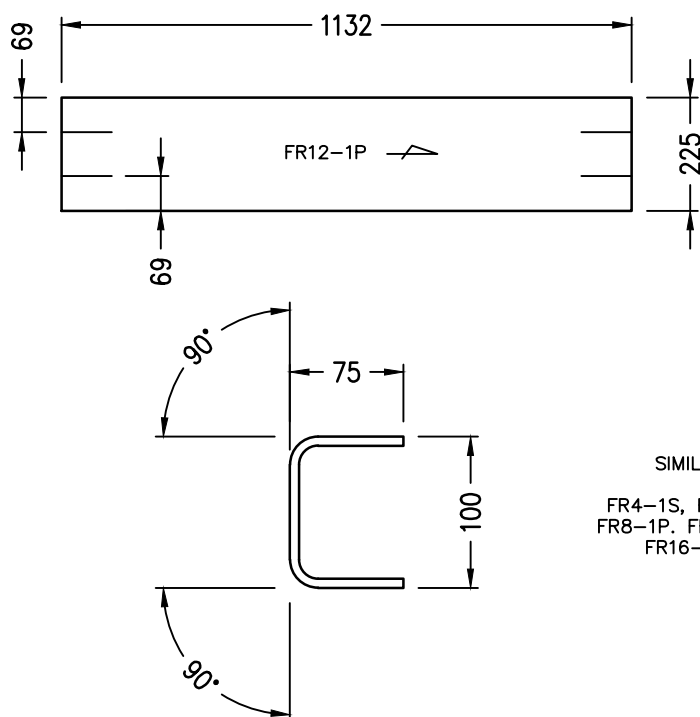
PART PRESSING DETAIL

PART #	VARIED
QTY	1
PRESS BAR DIA	Ø25
THICKNESS	6.00



SIMILAR FRAMES:

FR1-1C, FR2-1C, FR3-1C,
FR5-1C, FR6-1C, FR7-1C,
FR9-1C, FR10-1C, FR11-1C,
FR13-1C, FR14-1C, FR15-1C,
FR17-1C, FR18-1C, FR19-1C



SIMILAR FRAMES:

FR4-1S, FR4-1P, FR8-1S,
FR8-1P, FR12-1S, FR12-1P,
FR16-1S, FR16-1P

DO NOT SCALE

General Tolerances

LIN. ± 1 ANG. $\pm 2^\circ$

Checked

KIT ID / Proj. No.



NAVAL ARCHITECTS - MARINE CONSULTANTS - KIT SYSTEMS

Project

AFL SERVICES PLATFORM

DIMENSIONS IN mm

Scale

1:15

Sheet No.

1 OF 1

Issue

A